

BISHES RAYAMAJHI, EIT

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OBJECTIVE

More than four years of experience in Civil Engineering, Water Resources, Fish Passage, GIS, CAD design, Database Management, Computer Modeling, Scientific Research and Development. Seeking a full time job where I can apply my education, experience, and analytical skills to solve practical problems.

SKILLS

INSTRUMENTS : GPS, Theodolite, Real Time Kinematic (RTK), Oscilloscope and Stepper motors
PROCESSING : Hydraulics, Sediment transport, Water Quality, Topography, GIS, Image, Signal and Acoustics data
PROGRAMMING : Data acquisition, Automation, Motion control, Numerical modeling, Analysis and GUI development
ANALYTICS : Strong qualitative research, Statistical Inference, Complex and Large Data analysis
COMMUNICATION : Technical writings, Presentations, Team-player and Project Coordinator

SOFTWARE PROFICIENCY

Microsoft Office, AutoCAD 2D & 3D, ArcGIS, HECRAS, StreamStats, FishXing, ImageJ, Access, SQL, JMP, Sigma Plot, Tecplot, Total Commander, LabView and Agilent Waveform Builder Pro.

PROGRAMMING LANGUAGES

MATLAB, SAS, R, Python, C++, JAVA, FORTRAN, OCTAVE, Visual Basic and VHDL.

PROFESSIONAL EXPERIENCE

Pacific Northwest National Laboratory (US Department of Energy), North Bonneville, WA

Post Masters Research Associate

Jul 2012 – Present

- Designed and conducted lab tests for improving signal processing algorithms and real time fish detection software development.
- Extensively used ArcGIS for creating digital elevation models, water surface elevation profiles, temperature profiles and instrument location information.
- Created complex three-dimensional plots using AutoCAD, Sigma Plot and MATLAB for data analysis and design modeling.
- Automated data processing, analysis and QAQC by developing advanced algorithms in SAS and MATLAB.
- Developed and analyzed fish passage database (73,549 fish) from multiple years (2008 – 2012) of Juvenile Salmon Acoustic Telemetry System (JSATS) study.
- Involved in statistical analyses, survival prediction model development, research presentations and technical writings.

Maters Research Intern

Apr 2012 – Jul 2012

- Managed, processed and analyzed peta-bytes of water quality and hydro-acoustics data.
- Conducted hydrophone efficiency tests using remote boats and calculated positional errors.
- Deployed JSATS gears at several monitoring stations in the Columbia River basin.
- Regularly visited monitoring stations for their maintenance and well operation.

RELEVANT EXPERIENCE

Youngstown State University, Youngstown, OH

Aug 2010 – Apr 2012

Graduate Student Researcher

- Assisted in Civil Engineering undergraduate level course works, lab and grading assignments.
- Developed HEC-RAS models for simulating sediment transport and hydraulic effects due to dam removal.
- Estimated sediment transport volume and its impact zone due to low head dam removals.
- Developed a rating system for ranking low head dams in the Lower Mahoning River for their removal as a graduate thesis research.
- Assisted Ohio Environmental Protection Agency (OEPA) and local authorities involved in Mahoning River low head dam removals' planning and decision-making.

Summer Research Intern

- Determined the percentage of culverts in NE Ohio that were obstructing fish passage.
- Developed and simulated culvert fish passage using FishXing and HEC-RAS.
- Used StreamStats, National Elevation Dataset (NED) and ArcGIS for acquiring simulation model data.
- Evaluated the efficiency and user compatibility of the two software (HEC-RAS and FishXing) for fish passage.
- Published and presented research findings at World Environmental and Water Resources Congress, New Mexico, May 2012.

CE Construction Pvt. Ltd., Kathmandu, Nepal

Dec 2009 – May 2010

Construction Engineer

- Surveyed high-rise building foundation layout.
- Assured quality control on concrete, steel and formwork.
- Managed and supervised staffs and prepared their daily work schedules.
- Regulated proper safety measures.
- Periodically prepared quantity estimates of several construction materials.

EDUCATION

YOUNGSTOWN STATE UNIVERSITY, Youngstown, OH

M.S. Civil and Environmental Engineering

Overall GPA 3.9/4.0

May 2012

TRIBHUVAN UNIVERSITY, Pokhara, Nepal

B.S. Engineering in Civil Engineering

Overall GPA 3.1/4.0

July 2009

TRAININGS

AutoCAD 2D Training Course, Pokhara, Nepal

Building Interactive Applications (GUI) in MATLAB, MathWorks, Kirkland, WA

An Introduction to Interactive Programming in Python, Coursera

R Programming, Coursera

Data analysis and Statistical Inference, Coursera

CERTIFICATIONS AND MEMBERSHIP

Associate Member - American Society of Civil Engineers (ASCE)

2012 – Present

Engineer In Training (EIT) - State of Washington (CERT No. 33200)

2013

Registered Engineer - Nepal Engineering Council

2009

PUBLICATIONS*CONFERENCE PROCEEDINGS*

B Rayamajhi, JG Vasconcelos, JP Devkota, D Baral and HM Tritico. 2012. Should Fish Passage through Culverts Be a Concern for Midwest Engineers and Planners? Determining the Percentage of Culverts That Act As Barriers to Fish Passage in NE Ohio. World Environmental and Water Resources Congress 2012: Crossing Boundaries: pp. 1624-1634.

JP Devkota, D Baral, **B Rayamajhi**, and HM Tritico. 2012. Variation in Manning's Roughness Coefficient with Diameter, Discharge, and Slope in Partially Filled HDPE Culverts. World Environmental and Water Resources Congress 2012: pp. 1716-1726.

PEER-REVIEWED REPORTS

MA Weiland, JA Kim, **B Rayamajhi**, CM Woodley, GR Ploskey, J Rerecich and MB Eppard. 2015. Systematic Review of JSATS Passage and Survival Data at Bonneville Dam during Alternate Turbine and Spillbay Operations from 2008-2012. Pacific Northwest National Laboratory, Richland, WA.

B Rayamajhi, GR Ploskey, CM Woodley, MA Weiland, DM Faber, JA Kim, AHA Colotelo, Z Deng, and T Fu. 2013. Route-Specific Passage and Survival of Steelhead Kelts at The Dalles and Bonneville Dams, 2012. Pacific Northwest National Laboratory, Richland, WA.

RELEVANT COURSEWORKS

Advanced Engineering Mathematics, Environmental Engineering Design, Sediment Transport, Engineering Hydrology, Water Policy and Planning, Hydraulics, Fluid Mechanics, Numerical Methods, FORTRAN, Statistics, Hydropower, Surveying, GIS I & II.